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Evidence for deuteric magnetization in hydrothermally altered Mesozoic basaltic rocks from East Antarctica

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Mesozoic basaltic lavas and dykes from Vestfjella, East Antarctica, are dominated by well-grouped, single-component palaeomagnetic directions of normal polarity. Reversed magnetizations, heavily overprinted by normal polarity components, were encountered in a few lava flows and dykes. Only thermal demagnetization was successful in separating tight distributions of reversed polarity directions, interpreted to represent a deuteric magnetization. The region has been exposed to regional hydrothermal alteration to the epidote/prehnite metamorphic facies ($T \approx 300\,^{\circ}$C). Titanomagnetite grains in high-temperature oxidation classes II–III show alteration features typical for hydrothermally altered basalts; partial to complete replacement of both ilmenite and magnetite to sphene is common in the most altered rocks, which also exhibit varying degrees of decomposition of the opaque minerals. The lavas display two major, but distinctly different, thermomagnetic curves; the 'kink' type, previously reported from rocks exposed to the epidote metamorphic facies, and curves dominated by a paramagnetic contribution defining a magnetite Curie point. It is concluded that the basalts have retained palaeomagnetic directions acquired during the deuteric cooling phase, implying that granulation/sphene formation and Fe depletion of decomposed titanomagnetite grains in classes II–III do not necessarily cause complete remagnetization or the acquisition of significant secondary remanent magnetizations.

1. Introduction

The evaluation of stable palaeomagnetic directions relies on the assignment of proper ages of the directional components in relation to the age of the rock body in question. The deuteric magnetization acquired during the initial cooling of basaltic rocks is carried by titanomagnetites, TM(x) ($Fe_{3-x}Ti_xO_4$, $1 < x < 0$), invariably exposed to a range of deuteric as well as post-deuteric alterations. The palaeomagnetic response to post-deuteric alterations of TM, taking place at ambient to intermediate temperatures, is probably highly dependent on the spatial variation in deuteric alteration states of the TM grains. Low-temperature oxidation of TM in oxidation class I (Haggerty, 1976) results in the formation of cation-deficient spinels (maghemitization), accompanied by significant changes in both Curie temperature (T_c), saturation magnetization and the crystalline unit cell size. The degree of single-phase oxidation of TM may be expressed by the oxidation parameter z, ($1 < z < 0$), describing the increase in Fe^{3+} by oxidation of Fe^{2+}. The effect of maghemitization may cause time-dependent decreasing NRM intensities, explaining the apparent transient decrease in amplitudes of marine magnetic anomaly patterns with age (Bleil and Petersen, 1983). The effect of maghemitization upon the directional remanence properties, however, is still not clear, although experiments with natural and synthetic single-phase TM ($x \approx 0.6$) suggest that initial remanence directions are not influenced

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by moderate degrees of low-temperature oxidation ($z < \approx 0.6$) (Özdemir and Dunlop, 1985). Above $z \approx 0.6$, however, rhombohedral phases (hematite) may exsolve by phase splitting, causing the acquisition of a chemical remanent magnetization (CRM) parallel to the ambient field (Özdemir and Dunlop, 1985). This multicomponent magnetization is likely to be resolved by proper thermal demagnetization, although there is evidence to the contrary (Walderhaug and Torsvik, 1987). Comparable laboratory observations regarding the magnetic response of exsolved TM in deuteric oxidation states II and III is lacking. The magnetic and opaque mineralogical effects of hydrothermal alteration of basaltic rocks are quite significant. It has nevertheless been proposed that the deuteric magnetization would be retained even after exposure to the epidote metamorphic facies ($T > 250\,^{\circ}$C) (Ade-Hall et al., 1971). This conclusion was based on the supposition that if the observed widespread hydrothermal alterations of basaltic rocks also implied significant remagnetization, basaltic rocks would not in general yield consistent palaeomagnetic results.

The present paper reports on palaeomagnetic evidence for the apparent retention of two-polarity deuteric magnetization in basaltic rocks exposed to regional hydrothermal alteration in the prehnite/epidote metamorphic facies.

2. Geological setting and sampling

Vestfjella (73.74° S, 14.86° W) consists of small nunataks constituting the northermost part of the Transantarctic Mountains towards the Atlantic. Within a region of some 100×40 km, nunataks cover $< 2\%$ of the area (Fig. 1). They consist of subhorizontal tholeiitic lava flows, with thicknesses ranging between 1 and 10 m, with a re-

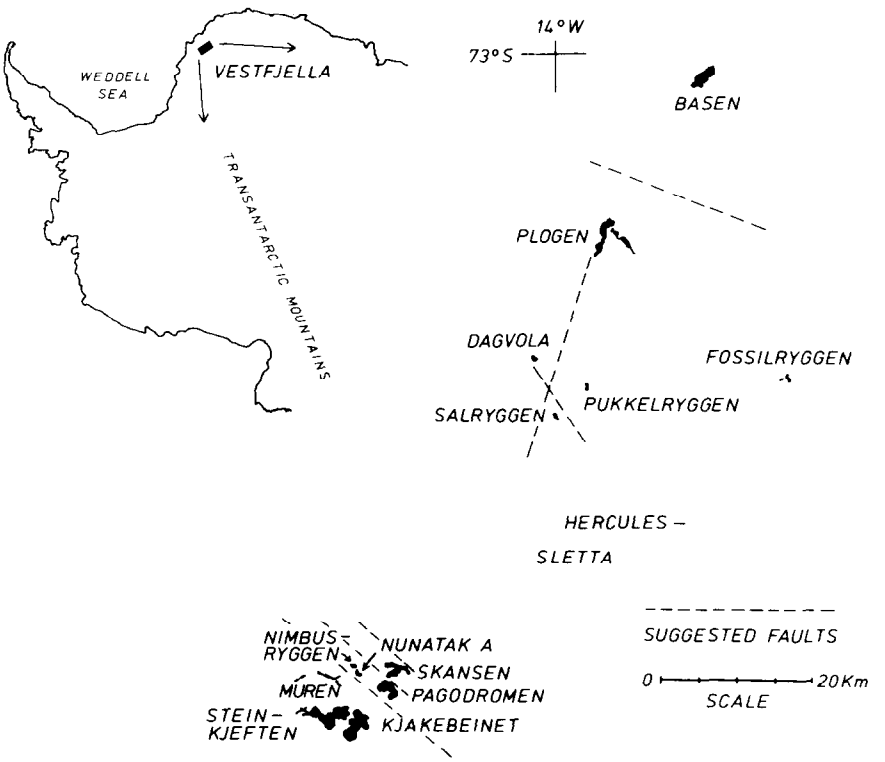


Fig. 1. Map showing basaltic nunataks from Vestfjella, Dronning Maud Land, East Antarctica, from which palaeomagnetic samples have been obtained.

gional dip almost due east. Locally, however, lava flows may tilt up to 15° towards N–S. The almost complete absence of interbedded sediments suggests a rapid and continuous rate of accumulation. The nunataks are cut by two suites of more or less vertical dykes, striking NE–SW and NW–SE, of which the former group is the younger, representing some 5% lateral extension of the region (Spaeth, 1986). The dykes are often associated with small-scale normal faults, down-throwing to the west as indicated by pronounced slickenside structures.

Radiometric age determinations of the dykes reveal consistent K–Ar ages ranging between 152 and 176 Ma (Furnes and Mitchell, 1978), as opposed to the lava flows which yield ages ranging from low values ≈ 170 Ma to unlikely high ages above 600 Ma. Although the spread in age determinations of the lavas nevertheless show a 'grouping' between 200 and 280 Ma, the evidently inconsistent results are attributed to the effect of regional hydrothermal activity, not necessarily pre-dating the intrusive event around 170 Ma. Oriented samples of dykes and lava flows, mostly in the form of 19 mm diameter drill cores, were collected during the Norwegian Antarctic Research Expeditions (NARE) in 1976–1977 and 1984–1985. The palaeomagnetic properties of the dykes, based on samples obtained during NARE 76/77, have been reported previously (Løvlie, 1979).

3. Evidence for regional hydrothermal activity

The petrological response of regional hydrothermal activity of basalts is characterized by alteration of primary minerals as well as by precipitation of secondary siliceous minerals (zeolites). The occurrence of characteristic secondary mineral assemblages reflects specific pressure–temperature stability fields, and may indicate maximum temperatures and depths of burial if the geothermal gradient can be estimated.

3.1. Secondary minerals

In the Vestfjella nunataks, assemblages of epidote and prehnite occur both in amygdales and pipe vesicle fillings, as well as on a much finer scale reflected by the spatially varying light to deep green colours of individual lava flows. The stability fields for epidote/prehnite suggest depth/ temperature conditions of around 2–3 kbar/ ≈ 300 °C (Ade-Hall et al., 1971).

3.2. Opaque mineralogy

The TM in both lavas and dykes occur as euhedral to anhedral grains (< 150 μm) in high-temperature oxidation class I–III (Haggerty, 1976). Granulation features are common in Class I and II grains, the former also showing pronounced curved intragrain cracks, indicative of titanomaghemite contraction cracks (Akimoto et al., 1984) (Fig. 2A). For grains in class II and higher, cracks are rare to absent, granulation features sparse, and white, bright shining titanohematite is more abundant. In general, dykes and lavas contain similar assemblages of TM phases. However, lava flows from some areas contain only relicts of TM in classes II–III, identified by the cubic shapes and trellis patterns of low-reflecting, translucent assemblages of sphene (?) and related secondary minerals (Fig. 2B). More often, however, ilmenite lamellae are completely altered to sphene, retaining the deuteric subdivision of the exsolved magnetite (Fig. 2C). Sphene also occurs in a more unusual mode (Haggerty, 1976), as a replacement of exsolved magnetite in TM class II–III (Fig. 2D). The green-coloured lavas contain almost no visible opaques, and the few observed grains (< 5 μm) are all embedded in a grey–white, translucent groundmass.

The groundmass of lavas containing partly decomposed TM grains is typically light grey to white, indicating that Fe dissolved from TM grains either did not precipitate in a ferric state, which is likely to produce a reddish-stained groundmass, or that Fe has been completely removed from the rock matrix. Ilmenite appearing as individual grains is usually more homogeneous and euhedral in shape, showing less-developed alteration features due to regional hydrothermal metamorphism. Minor amounts of sulphides (pyrite) are only encountered in the dykes.

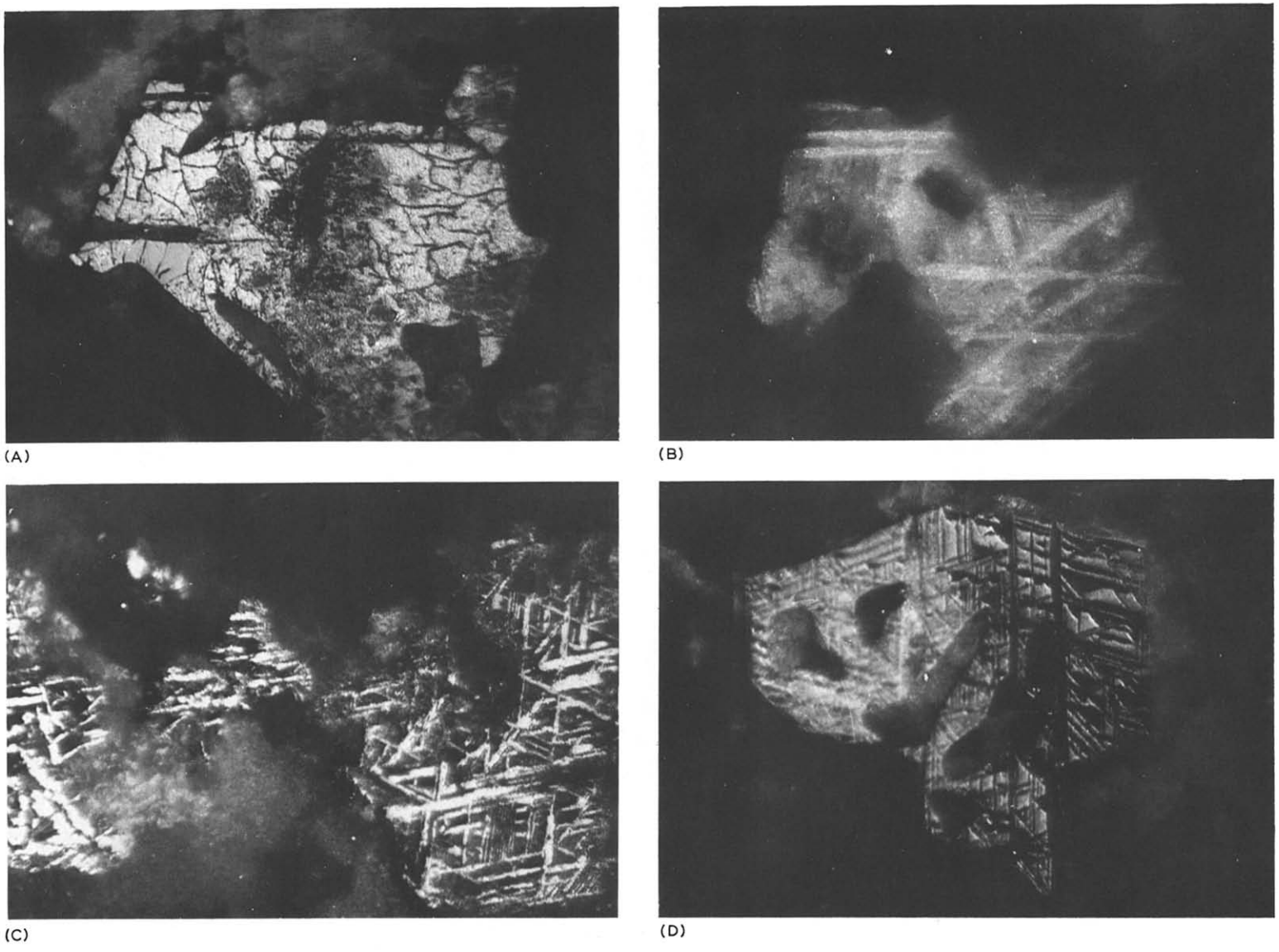


Fig. 2. Photomicrographs of titanomagnetite (TM) grains displaying characteristic hydrothermal alteration features. (A) Intragrain contraction cracks. (B) Relict of TM grain completely altered to sphene and non-opaque mineral phases. (C) TM grain with ilmenite lamellae completely altered to sphene. (D) Magnetite replaced by sphene (?) associated with unaltered ilmenite lamellae.

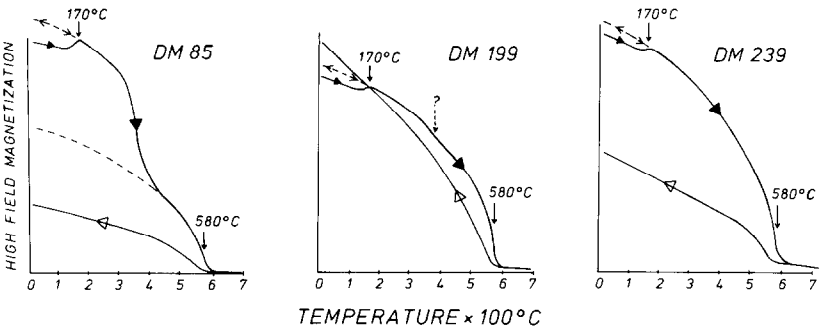


Fig. 3. Typical thermomagnetic curves of type A, exhibiting the characteristic 'kink' at 170 °C. Heating/cooling curves indicated by filled/open arrows. Broken curves: cooling/heating from intermediate temperatures. Note different shapes of heating curves above 200 °C, which nevertheless all define a high-temperature Curie point around 580 °C.

3.3. *Thermomagnetic properties*

The magnetic response of basalts to regional hydrothermal alteration has previously been classified by a set of thermomagnetic curves with features characteristic for different zeolite metamorphic facies (Ade-Hall et al., 1971). Thermomagnetic curves of the Vestfjella basalts were obtained by an automatic recording, transversal Curie balance in fields ranging between 0.4 and 0.6 T. All experiments were performed in air with heating/cooling rates between 15 and 30 °C min^{-1}. The thermomagnetic properties are dominated by two distinctly different curve types, of which type A (Fig. 3) exhibits features typical for basalts in deuteric oxidation class higher than I and exposed to epidote metamorphism (Ade-Hall et al., 1971). This dominating group is characterized by a more or less pronounced increase in saturation magnetization (J_s) at 170 °C, identical to the kink reported by Ade-Hall et al. (1971) to occur in the 100–200 °C range. Subsequent cooling from 200 °C demonstrates that the increase in high-field magnetization represents the irreversible creation of a magnetic phase with a higher saturation magnetization than the initial phase. Thermomagnetic runs of finely ground rock ($\approx$ 10–20 μm) and larger chips ($\approx$ 2 mm) yield similar curves, suggesting that this process is not very sensitive to O_2 supply and may thus represent some kind of structural inversion process. This observation suggests that the rocks may not have been heated to this temperature range in nature since the formation of this phase.

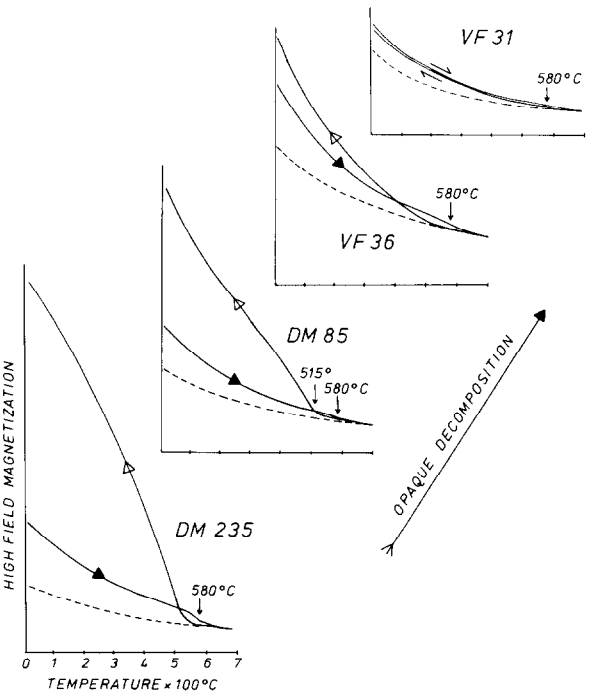


Fig. 4. Set of typical thermomagnetic curves of type B for TM grains in deuteric oxidation classes II–III. Heating/cooling curves indicated by filled/open arrows. The curves represent different degrees of visual opaque decomposition, increasing from lower left to upper right. Note the apparent systematic decrease in J_f/J_i ratios with increasing stage of dissolution and the significant paramagnetic contribution at 700 °C. Broken curve: induced paramagnetic magnetization.

The concave part of the curve between 300 and 400 °C is thermomagnetically irreversible, demonstrated by cooling from 450 °C, signifying the inversion of a maghemite phase to a weaker magnetic phase. Heating curves define a T_c around 580 °C, indicative of magnetite, which upon cooling from 700 °C results in a slightly, but significantly, lower T_c. All A-type curves represent TM in deuteric oxidation classes I–II.

The other type of curve (B) (Fig. 4), is more common in the Vestfjella basalts, and was not reported by Ade-Hall et al. (1971) as typical for hydrothermally altered basalts. Concave heating curves, with sometimes weakly defined T_c around 580 °C, are always associated with a significant paramagnetic contribution above 660 °C. Heating/cooling to 600 °C yields reversible thermomagnetic curves. Upon cooling from 700 °C, however, a much stronger magnetic phase is created, associated with $T_c \approx 560$ °C. Although this secondary phase is likely to represent almost pure magnetite, it probably did not form by the inversion process of titanomaghemite phases, which is known to take place below 600 °C (O'Reilly, 1983). The process may either represent magnetite formation created from non-opaque Fe sources (clays?), or TM exsolution. The ratio of final to initial high-field magnetization, J_f/J_i, may reach 7, but is typically between 2 and 3.

Samples with thermomagnetic curve type B all contain partly dissolved TM class II–III grains, with sphene.

4. Palaeomagnetic properties

The Vestfjella basaltic dykes have previously been reported to carry predominantly normal polarity palaeomagnetic directions of inferred Jurassic age (Løvlie, 1979). All normal polarity

TABLE I
Palaeomagnetic site mean directions Vestfjella

Nunatak	Lava flows						Dyke intrusions					
	N	*(N)*	*D*	*I*	*k*	α_{95}	*N*	*(N)*	*D*	*I*	*k*	α_{95}
Normal polarity												
Pukkelryggen	3	(16)	30	−53	87	8.7	2	(13)	43	−60	965	3.2
Salryggen	8	(35)	31	−61	144	4.1	1	(7)	17	−51	(189)	(3.9)
Skansen	4	(15)	31	−55	27	13.1	4	(20)	42	−68	43	10.8
Pagodromen NW	13	(41)	39	−61	45	5.8	3	(15)	25	−61	98	8.2
Nimbusryggen	4	(14)	14	−65	530	3.0	1	(6)	47	−44	(15)	(17.0)
Muren	11	(36)	8	−62	58	5.6	2	(11)	6	−55	12	28.7
Steinkjeften	3	(8)	26	−59	97	8.2	4	(14)	33	−60	33	12.2
Kjakebeinet	7	(20)	31	−56	139	4.5	3	(11)	35	−56	80	9.1
Nunatak A	1	(4)	21	−74	(1836)	(2.2)	4	(19)	28	−58	232	4.6
Basen	3	(6)	28	−40	129	5.0						
	1	(4)	54	−76	95	7.2						
Grushallet	2	(10)	38	−47	360	5.2						
Dagvola	3	(14)	353	−59	345	4.4						
Fossilryggen							3	(9)	20	−40	206	5.6
							1	(7)	21	−84	(63)	(6.6)
Mean	13		26	−60	46	5.7	11		29	−58	36	7.0
Reversed polarity												
Plogen	6	(23)	203	36	171	4.4						
Pagodromen SE	4	(18)	220	54	184	5.2						
Nimbusryggen							1	(4)	209	56	(331)	(5.1)
Skansen							2	(9)	226	51	105	9.6
Basen							1	(4)	222	38	(20)	(16.0)
Mean	2		210	45			3		220	49	56	10.8

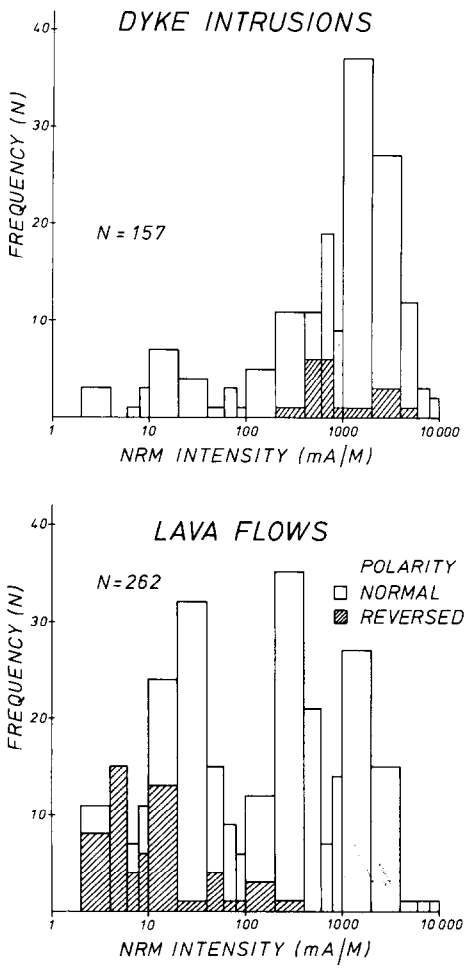


Fig. 5. Distribution of NRM intensities for lavas and dykes.

rocks carried single-component magnetizations, as opposed to the reversed rocks which are overprinted by significant normal magnetic components (Løvlie, 1979). The present results, mainly from lava flows, are in general accord with these findings. A summary of the palaeomagnetic results from Vestfjella is given in Table I.

The normal polarity dykes have NRM intensities ranging between 2.10^{-3} and 10 A m^{-1}, with > 50% above 1 A m^{-1}. The lavas define a similar range but have a significantly broader distribution of NRM intensities (Fig. 5). The reversed polarity lavas all have the lowest NRM intensities. The distributions of median destructive fields (MDF) indicate a reverse relationship; the lavas, irrespective of polarity, appear to have slightly higher MDFs than the dykes. In both rock types, however, MDFs > 80 mT are always associated with thermomagnetic curves of type A, which reflects highly granulated TM grains. Granulation evidently caused an extremely fine subdivision of magnetic grains, giving rise to the high coercivities.

Subsequent thermal demagnetization of samples with very high magnetic stability against alternating field (af) treatment (NRM intensity reductions of the order of < 20% after 80 mT), reveal single magnetic components with major unblocking above 300 °C (Fig. 6). Alternating field and thermal demagnetization of normal polarity samples with type B thermomagnetic curves also revealed single-component magnetizations associated with significantly lower MDFs and major unblocking above 450 °C (Fig. 7).

Multicomponent magnetizations were encountered in only two lava flow successions (SE Pagodromen and Plogen) and in three dykes (Nimbusryggen, Skansen and Basen). Alternating field demagnetization was only partly successful in isolating reversed directions, consistently overprinted by normal polarity components. Thermal demagnetization to > 400 °C, however, generally succeeded in removing normal polarity overprints (cf., DM205-A1, Fig. 8).

5. Evidence for deuteric magnetization

5.1. Magnetostratigraphic correlation

Palaeomagnetic mean declinations from 15 of the most fresh-looking lava flows along the NW Pagodromen display consistent variations defining more than two cyclic periods with a maximum amplitude of around 70°. These large deviations in declination cannot be ascribed to local tectonics within the ca. 400 m thick pile of subhorizontal lava flows. Similar unusually large easterly mean directions were also encountered in lava flows from the nearby Skansen nunatak (≈ 2.5 km towards the NW), showing a consistent stratigraphic pattern of declination changes that may be correlated with the Pagodromen section, as suggested in

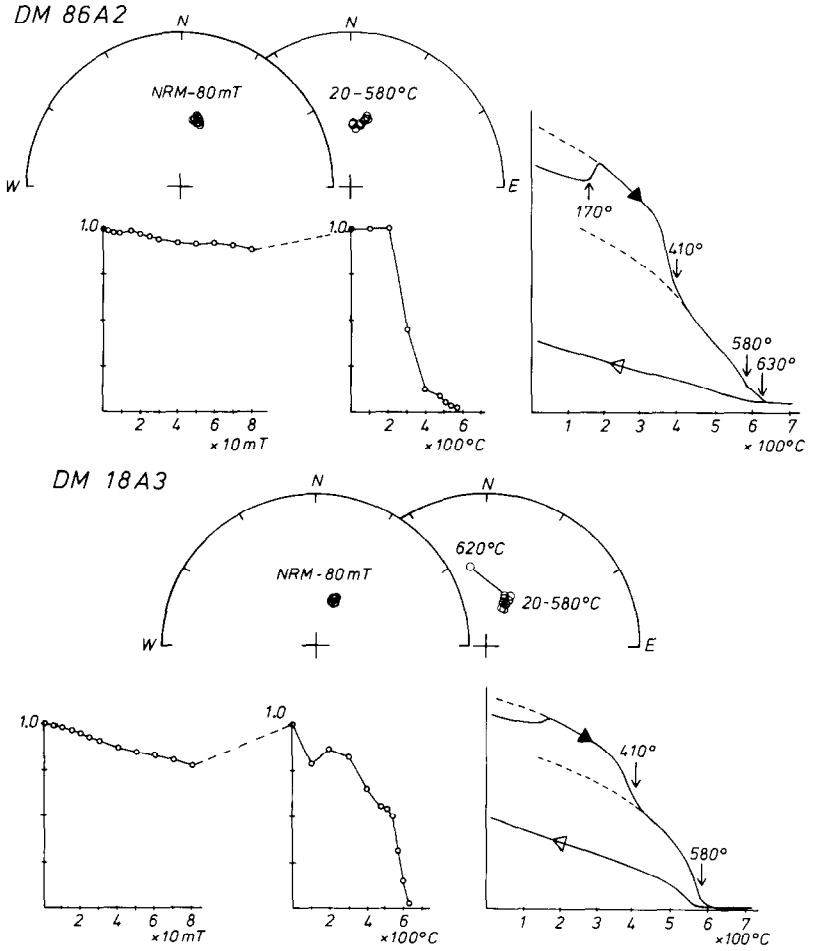


Fig. 6. Composite demagnetization results of lava samples with single-component magnetization and extremely high remanence coercivities (MDF > 80 mT), always associated with thermomagnetic curves of type A, shown to the right. Note that almost identical thermomagnetic curves are not reflected by corresponding similarity of the thermal decay curves: DM86A2 exhibits a major decrease in magnetization between 300 and 400 ° C, while unblocking of DM18A3 takes place above 500 ° C.

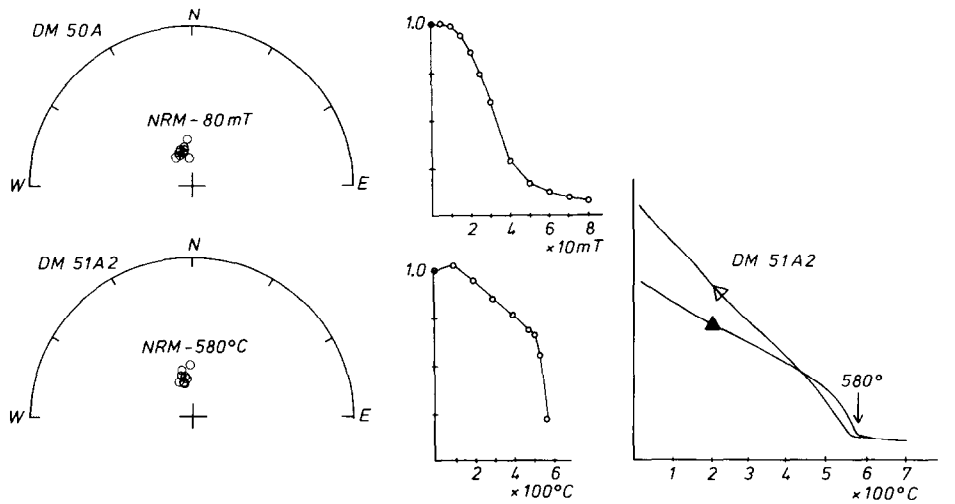


Fig. 7. Alternating field and thermal demagnetization results from two nearby samples from the same lava flow (Dagvola), with thermomagnetic curves of type B, shown to the right.

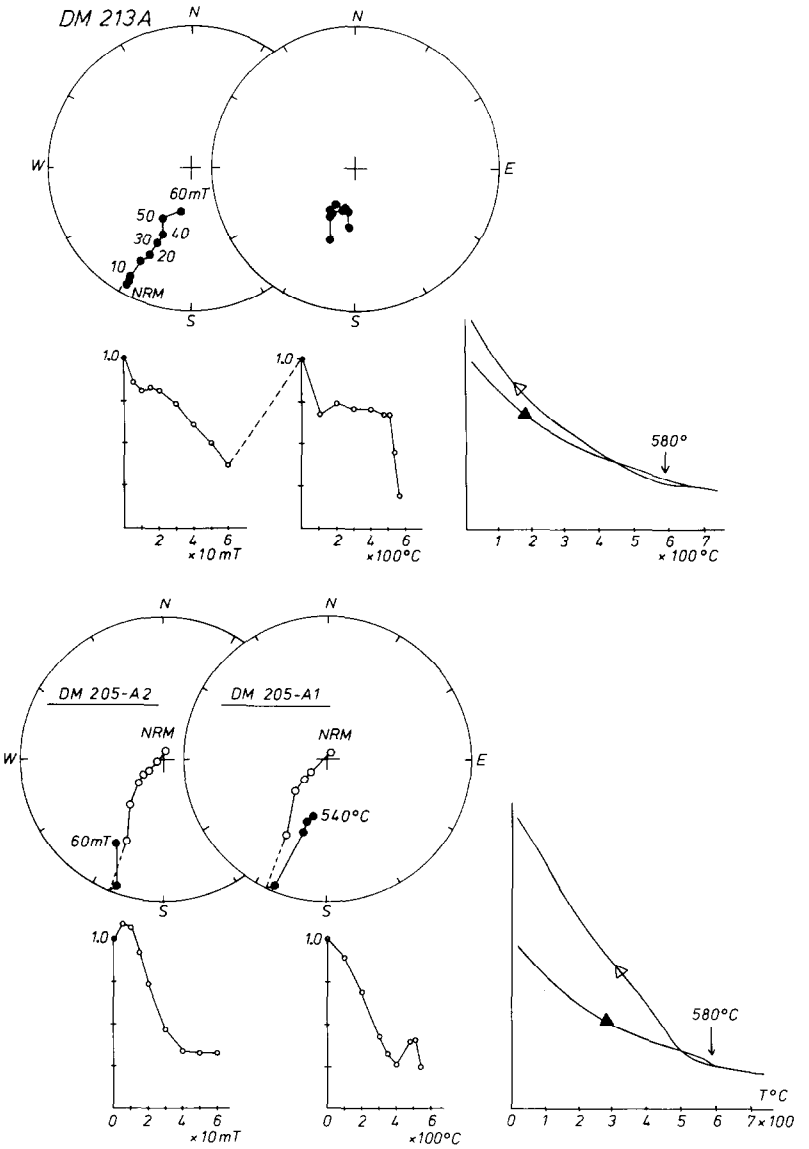


Fig. 8. Alternating field and thermal demagnetization results of samples with thermomagnetic curves of type B (shown to the right) from two reversed-polarity lava flows from Plogen. The normal component overprint in sample DRM213A (top) is successfully removed by af demagnetization, as demonstrated by the closely grouped directions obtained by subsequent thermal demagnetization. More typically, composite components remained even after treatment to 60 mT, cf., specimen DM205-A2 (bottom). Single-component reversed magnetizations are associated with blocking temperatures above 400 ° C, cf., specimen DM205-Al (bottom).

Fig. 9. The correlation is evidently ambiguous, since the Skansen pattern may also fit in at the top of the Pagodromen section (Fig. 9). Similar pronounced cyclic features are absent in the inclination data, preventing an independent magnetostratigraphic correlation. Ice cover precludes visual stratigraphic correlation of individual lava flows between Pagodromen and Skansen, and the proposed correlation is therefore based on palaeomagnetic mean directions only. Considering the congruous patterns of anomalously large amplitude variations in declinations present in

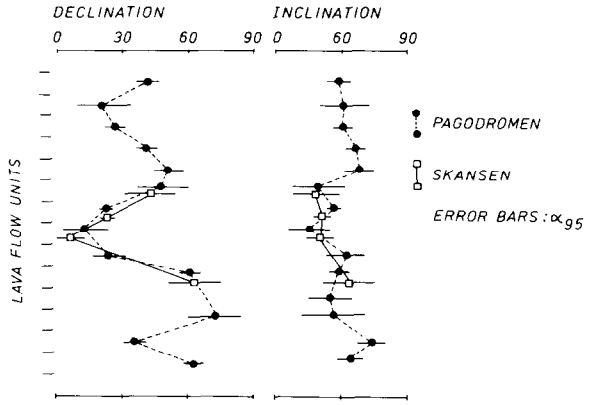


Fig. 9. Magnetostratographic correlation based on mean declinations of individual lava flows from Pagodromen (●) and Skansen (□). The height scale represents the number of successive fresh-looking lava flows at the two nunataks.

successions of lava flows at lateral distances > 2 km, the apparent good stratigraphic agreement between mean directions is regarded as strong evidence that the lava flows in question carry concurrent records of the geomagnetic field.

5.2. *Magnetopolarity*

Multicomponent palaeomagnetic directions in lavas at SE Pagodromen and Plogen define four discrete magnetization components (A–D), confined within discrete blocking temperature intervals (Fig. 10). The components were separated by a detailed analysis of optimal orthogonal vector plots of thermal demagnetization results, utilizing a linear regression-based, line-find program (Torsvik, 1987).

The scattered directional distribution of the A components, defined by the NRM to 100 °C vector and representing < 10% of the total NRM, is interpreted to represent viscous magnetic overprints acquired after sampling.

The tight distribution of the B components (100–300 °C/400 °C), constituting some 15% of

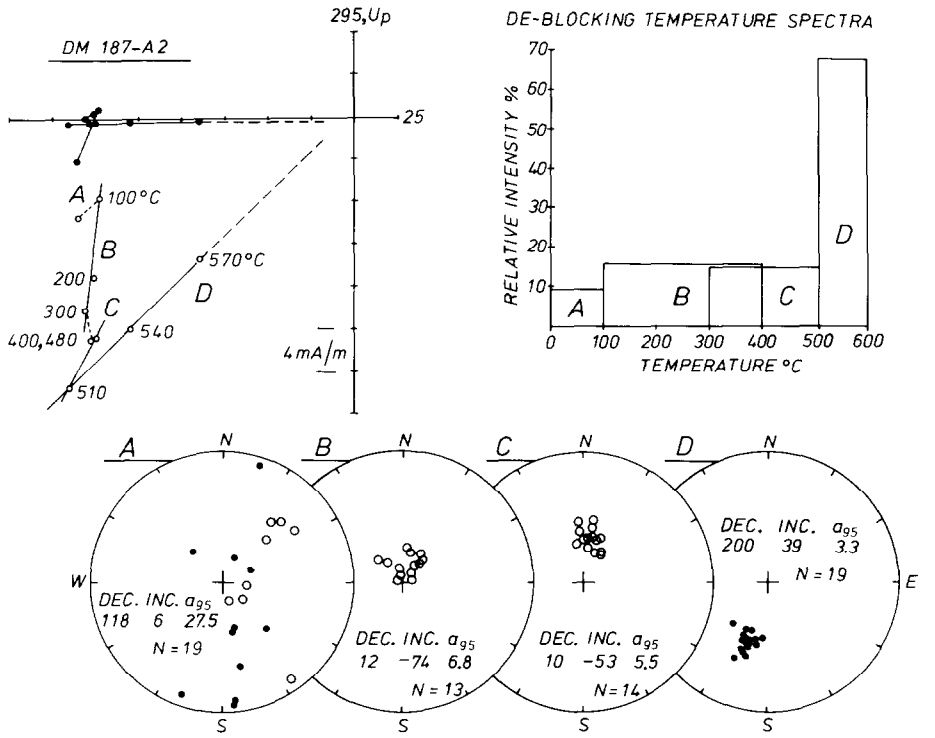


Fig. 10. Palaeomagnetic components separated by thermal demagnetization of lava flows from Plogen. Decomposition of NRM illustrated by the optimum orthogonal vector plot, upper left. Unblocking temperature spectra, upper right. Stereographic projections: directional distributions of components A–D, with Fisherian mean declination, inclination and angle of confidence, α_{95}.

the NRM, has inclinations comparable with those of the present geomagnetic field. This component may thus either reflect a viscous overprint (Quaternary) or be related to a late magmatic phase, which may also be represented by a dyke at the Fossilryggen nunatak, and by some lava flows above a pronounced discordance at Basen (cf., Table I).

The mean direction of the C components (300 °C/400–500 °C) coincides with the mean of the normal polarity rocks within the Vestfjella basalts. Hence this component is likely to represent an overprint reflecting the major magmatic episode associated with the emplacement of the normal polarity lavas and/or dykes. The line-find program discriminated between components B and C in only a few samples, and the presented distribution of these components was obtained by selective vector-subtraction in the appropriate temperature intervals. The palaeomagnetic significance of components B and C may thus be dubious.

The composite magnetization is dominated by the reversed D component, which is confined within a narrow blocking-temperature range (500– < 600 °C) and representing some 70% of the total NRM. The normal polarity B and C components, overprinting the reversed D component, are tentatively interpreted to reflect one, or possibly two, post-emplacement magmatic events. The low scatter of the reversed polarity directional distribution ($\alpha_{95} = 3.3°$) is concluded to indicate the successful separation of single-component palaeomagnetic directions.

The reversed polarity magnetizations in the SE Pagodromen and Plogen lavas define two tight distributions ($\alpha_{95} = 5.2°/4.4°$) with significantly different mean directions, attributed to either varying post-emplacement tilting of the lava flows, or to age differences of the magnetizations (cf., Table I).

6. Discussion

6.1. Origin of magnetization

The apparent single-component palaeomagnetic directions residing in the normal polarity lavas and dykes may be attributed to three different modes of origin: (1) the basaltic rocks (lavas and dykes) were formed during a time interval of almost stationary palaeomagnetic pole, (2) the lavas were completely remagnetized at the time of dyke intrusion, or (3) both lavas and dykes were completely remagnetized by some post-intrusive event. One likely candidate for the last mode is the final fragmentation of East Antarctica from Africa, in Mid-Kimmeridgian time and pre-dating marine magnetic anomaly M21 ≈ 145 Ma (Martin and Hartnady, 1986). However, continental break-up by crustal separation appears to begin at, or shortly after, a peak in igneous activity (Scrutton, 1973). The onset of magmatism may precede the formation of new oceanic crust by 20–25 Ma (Scrutton, 1973), and the dykes in Vestfjella (≈ 150–180 Ma) are thus likely to represent the last magmatic event in this region.

Lava flows with heavily overprinted reversed polarity palaeomagnetic directions preclude complete remagnetization in a normal polarity field, implying that the last intrusive phase did not cause a regional resetting of the deuteric magnetization in the inferred oldest lavas.

Likewise, the proposed magnetostratigraphic correlation, based on consistent patterns of large amplitude variations of normal polarity mean declinations from individual lava flows from Pagodromen and Skansen (cf., Fig. 8), does not give supporting evidence for partial remagnetization associated with post-emplacement magmatic events.

6.2. Titanomagnetite alterations

The present observations suggest that TM in high-temperature oxidation classes II–III may respond to hydrothermal alteration in the epidote/prehnite stage by the formation of sphene/titanohematite, in addition to partial to complete decomposition of TM grains.

Titanomagnetite alterations may also be distinguished by characteristic features on thermomagnetic curves. The Vestfjella basalts are dominated by two distinctly different thermomagnetic curves. The B-type curves are characterized by low bulk saturation magnetizations, more or

less well-developed magnetite Curie points, and by significant induced paramagnetic magnetization. The latter may in part be due to a decreasing ferrimagnetic contribution caused by partial decomposition of TM grains. Fe dissolved in the hot groundwater probably did not precipitate as Fe^{3+} in the groundmass, as indicated by the grey to white colour of the latter. Although the Fe in solution may thus have been carried away by percolating groundwater, it is tentatively suggested that at least a fraction of the Fe precipitated in a Fe^{2+} state in some secondary mineral phases, giving rise to the enhanced paramagnetic magnetization (Fig. 3). It is proposed that the latter phases may also contribute to the thermally activated production of magnetite, observed to take place upon heating to above 600 °C (Fig. 3).

Thermomagnetic type A curves, with the typical kink below 200 °C do not display significant paramagnetic contributions above 700 °C, indicating insignificant Fe depletion from the TM grains in accordance with the absence of pronounced decomposition features of the TM grains. The groundmass in samples showing type A thermomagnetic curves is stained brown-red, indicative of Fe^{3+} migration, which is common for basalts exposed to hydrothermal alteration. There is no evidence in these samples for the precipitation of assemblages of secondary magnetic minerals (magnetite) which may carry a chemical remanent magnetization (Schoenharting and Ghisler, 1982), potentially indiscernible from a deuteric magnetization.

7. Conclusions

The Mesozoic Vestfjella basalts, exposed to regional hydrothermal metamorphism to the prehnite/epidote facies, are concluded to have retained palaeomagnetic directions acquired during the deuteric phase. This conclusion is based on observed large amplitude, coherent variations of inferred concurrent single component palaeomagnetic directions along two lava successions, irreconcilable with partial or complete remagnetization. Normal polarity components, overprinting the inferred oldest rocks carrying reversed polarity directions, may reflect the magmatic events during which the normal polarity lavas and dykes were emplaced. The remanence-carrying TM grains (classes I–III) show alteration features ranging from contraction cracks (maghemitization), granulation, titanohematite formation and partial to complete decomposition of TM grains. The retention of primary palaeomagnetic directions of reversed polarity, in spite of extensive TM alterations which took place in a normal polarity field, is inferred to be partly due to the absence of precipitated magnetic minerals. Magnetic subdivision and partial to complete decomposition of TM grains in deuteric oxidation classes II and III due to low-temperature alterations, may hence not cause the acquisition of secondary remanent magnetization. These findings substantiate the conclusion made by Ade-Hall et al. (1971) that suitable cleaning methods may yield the original cooling remanence directions in hydrothermally altered basaltic rocks.

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